Study Notes: Electromagnetic Theory and Vector Calculus for Electrical Engineers

Electromagnetic Theory and Vector Calculus

Engineering Physics

ACADEMIC LEVEL

btech · B.Tech Electrical Engineering

DIFFICULTY

beginner

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Prepared for learning and revision. Check course-specific requirements with your instructor.

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Learning Objectives

- Define scalar and vector fields and apply the del operator (gradient, divergence, and curl) in Cartesian coordinates.
- State and apply Gauss's Divergence Theorem and Stokes' Theorem to transform surface, volume, and line integrals.
- Formulate Maxwell's four equations in both differential and integral forms, explaining the physical significance of each.
- Explain the concept of Maxwell's displacement current and demonstrate why Ampÿýre's circuital law required modification for time-varying fields.
- Derive the electromagnetic wave equation in free space from Maxwell's equations and determine the propagation velocity of light.

1. Introduction to Vector Calculus in Electromagnetics

Electromagnetic theory describes the interactions of electric charges, electric fields, magnetic fields, and energy propagation. To formulate these physical concepts quantitatively, we use vector calculus.

A scalar field assigns a single scalar value (such as electrostatic potential

V

in volts,

V

) to every point in space. A vector field assigns a vector quantity (having both magnitude and direction, such as electric field

$\vec{E}$

in volts per meter,

[redacted]

V/m

[redacted]

or magnetic flux density

$\vec{B}$

in teslas,

T

) to every point in space.

2. The Del Operator and Vector Operations

The fundamental differential operator in vector calculus is the nabla or del operator, denoted by

∇.

In

$three-dimensional$

Cartesian coordinates

(x, y, z),

it is defined as:

$$\n\$abla = \hat{i}\$\$\frac{\partial}{\partial x} + \hat{j}\$\$\frac{\partial}{\partial y} + \hat{k}\$\frac{\partial}{\partial z}$$

Here,

$\hat{i}, \hat{j}, \hat{k}$

are unit vectors along the

$x, y,$

and

z

axes.

2.1 Gradient of a Scalar Field (nabla V)

- Physical meaning: The gradient points in the direction of the maximum spatial rate of increase of the scalar field, and its magnitude equals this maximum rate.
- Formula: For a scalar field V(x, y, z):

$$\nabla\$V = \hat{i}\$\$\frac{\partial V}{\partial x} + \hat{j}\$\$\frac{\partial V}{\partial y} + \hat{k}\$\frac{\partial V}{\partial z}$$

- In electrostatics, the electric field is the negative gradient of electric potential: vec(E) = -nabla V.

2.2 Divergence of a Vector Field (nabla * vec(A))

- Physical meaning: Divergence measures the net outward flux per unit volume from an infinitesimal region. A positive divergence signifies a source; a negative divergence signifies a sink; zero divergence signifies an incompressible or solenoidal field.
- Formula: For vec(A) = Axhati + Ayhatj + A_zhatk:

$$\n$abla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

2.3 Curl of a Vector Field (nabla x vec(A))

- Physical meaning: Curl measures the circulation or rotational tendency of a vector field per unit area around a point.
- Formula:

$$\n$abla \times \vec{A} = \begin{vmatrix}$$

3. Integral Theorems

1. Gauss's Divergence Theorem: Relates the volume integral of the divergence of a vector field over a closed volume V to the surface integral over the bounding closed surface S:

$$\iiint_V (\$\nabla \cdot \vec{A}\$)\, \$dV = \oint_S \$\$\vec{A} \cdot d\$\vec{S}$$

2. Stokes' Theorem: Relates the surface integral of the curl of a vector field over an open surface S to the line integral around the closed bounding contour C:

$$\iint_S (\$\nabla \times \vec{A}\$) \cdot d\$\vec{S} = \oint_C \$\$\vec{A} \cdot d\$\vec{r}$$

4. Maxwell's Equations

Maxwell unified electricity and magnetism into four core equations. They exist in both differential (point) form and integral form.

4.1 Gauss's Law for Electrostatics

- Differential form: nabla * vec(D) = rhov(or in linear, isotropic media:nabla * vec(E) = (rhov)/(varepsilon))
- Integral form: ointSvec(D) * dvec(S) = Qtextenclosed
- Meaning: Total electric flux leaving a closed volume equals the free electric charge enclosed.

4.2 Gauss's Law for Magnetism

- Differential form: nabla * vec(B) = 0
- Integral form: closed integral_Svec(B) * dvec(S) = 0

- Meaning: Magnetic field lines are continuous closed loops; isolated magnetic monopoles do not exist in classical electromagnetism.

4.3 Faraday's Law of Electromagnetic Induction

- Differential form: nabla x vec(E) = -fracpartial vec(B)partial t
- Integral form: ointCvec(E) * dvec(r) = -(d)/(dt)iintS vec(B) * dvec(S)
- Meaning: A time-varying magnetic field induces a circulation of electric field (an electromotive force).

4.4 Ampÿýre-Maxwell Law

- Differential form: nabla x vec(H) = vec(J) + fracpartial vec(D)partial t
- Integral form: ointCvec(H) * dvec(r) = iintS (vec(J) + fracpartial vec(D)partial t) * dvec(S)
- Meaning: Magnetic fields are produced both by conduction current density vec(J) and by time-varying electric displacement field fracpartial vec(D)partial t (displacement current density vec(J)_d).

5. Displacement Current and the Inconsistency of Ampÿýre's Original Law

Ampÿýre's original static law stated

$$\n$$

abla x vec(H) = vec(J). Taking the divergence of both sides:

$$\nabla \cdot (\$\nabla \times \vec{H}\$) = \$\nabla \cdot \vec{J}\$$$

Because the divergence of any curl is identically zero:

$$0 = \$\nabla \cdot \vec{J}\$$$

However, the law of conservation of charge is given by the continuity equation:

$$\n\$abla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}\$$$

This is consistent with

$$\n$$

abla * vec(J) = 0

$$only for steady states (where$$

(partial rho_v)/(partial t) = 0

$$). For$$

time-varying

$$fields, suchaschargingacapacitor, chargeaccumulatesonplates, making$$

(partial rho_v)/(partial t) != 0.

To resolve this contradiction, Maxwell substituted Gauss's Law (

$$\r$$

ho_v = nabla * vec(D)) into the continuity equation:

Therefore, the total

$$divergence - free$$

current density is

$$\vec{J}_{\text{total}} =$$

vec(J) + fracpartial vec(D)partial t, giving the corrected law:

$$\n\$abla \times \vec{H} = \vec{J} + \vec{J}_d, \$ \quad \text{where } \$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}\$$$

Key Terms

- Scalar Field: A mathematical function that associates a single real physical value (without spatial direction) to every point in a region of space.
- Vector Field: A mathematical function that assigns a vector (having magnitude and direction) to every coordinate point in a region of space.
- Divergence: A scalar differential operator measuring the net rate at which a vector field flows outward from an infinitesimal volume element.
- Curl: A vector differential operator measuring the tendency of a vector field to circulate or rotate around an axis passing through a point.
- Displacement Current Density: A quantity defined as the partial time derivative of the electric displacement field (?D/?t), which produces a magnetic field just as physical moving charges do.
- Permittivity (?): A measure of a material's ability to resist an electric field by polarizing internally; in vacuum, ?? ? 8.854 ÿý 10?ÿýÿý F/m.
- Permeability (?): A measure of a material's ability to support the formation of a magnetic field; in vacuum, ?? = 4? ÿý 10?? H/m.

Key Formulas

Divergence of a Vector Field

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Calculates the net flux divergence of vector field A = (Ax, Ay, Az) per unit volume in Cartesian coordinates. SI unit depends on the field (e.g.,

$$V/m^2$$

if A is electric field).

Curl of a Vector Field

$$\nabla \times \vec{A} = \hat{i}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) - \hat{j}\left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z}\right) + \hat{k}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

Measures spatial circulation per unit area in Cartesian coordinates.

Gauss's Divergence Theorem

$$\iiint_V (\nabla \cdot \vec{A})\, dV = \oint_S \vec{A} \cdot d\vec{S}$$

Converts a volume integral over closed region V to a surface integral over enclosing closed surface S.

Stokes' Theorem

$$\iint_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_C \vec{A} \cdot d\vec{r}$$

Converts a surface integral of curl over open surface S to a line integral around the bounding perimeter C.

Displacement Current Density

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \varepsilon \frac{\partial \vec{E}}{\partial t}$$

J_d is displacement current density (

$$A/m^2),$$

D is electric flux density (

C/m^2),

E is electric field (V/m), ? is medium permittivity (F/m).

Electromagnetic Wave Equation in Free Space

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Governs the propagation of electric field vectors in a $source - free$ $non - conducting$ vacuum (? = 0, $J = 0$). The wave speed is $c = 1/sqrt$ (????).

Derivations

Derivation of the Electromagnetic Wave Equation in Free Space

Step 1: Write Maxwell's equations in a linear, isotropic, homogeneous, $charge - free$ ($\$\rho_v = 0\$$), and $current - free$

(

$$\$\vec{J} = 0\$$$

) vacuum:

1) \(nabla * vec(E) = 0\)
2) \(nabla * vec(B) = 0\)
3) \(nabla x vec(E) = -fracpartial vec(B)partial t\)
4) \(nabla x vec(B) = mu0 varepsilon0 fracpartial vec(E)partial t\)

Step 2: Take the curl of Faraday's Law (Equation 3):

Step 3: Interchanging spatial and temporal partial derivatives on the

$$right - hand$$

side yields:

$$\nabla \times (\$\nabla \times \vec{E}\$) = -\frac{\partial}{\partial t}(\$\nabla \times \vec{B}\$)$$

Step 4: Use the standard vector identity for the double curl:

$$\nabla \times (\$\nabla \times \vec{E}\$) = \nabla(\$\nabla \cdot \vec{E}\$) - \nabla^2 \vec{E}.$$

Step 5: Apply Gauss's Law in free space (

$$\$\nabla \cdot \vec{E} = 0\$$$

). The gradient term vanishes:

$$\nabla(\$\nabla \cdot \vec{E}\$) - \nabla^2 \$\vec{E} = -\nabla^2 \$\vec{E}$$

Step 6: Substitute Equation (4),

$$\$\nabla \times \vec{B} = \mu_0 \$\varepsilon_0 \frac{\partial \vec{E}}{\partial t},$$

into the

$$right - hand$$

side:

$$-\nabla^2\$\vec{E} = -\frac{\partial}{\partial t}\$\left(\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}\right) = -\mu_0\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$$

Step 7: Cancel the negative sign to obtain the 3D wave equation:

$$\nabla^2\$\vec{E} = \mu_0\$\varepsilon_0\frac{\partial^2\vec{E}}{\partial t^2}$$

Comparing this with the classical 3D wave equation

$$\nabla^2\$\psi = \frac{1}{v^2}\$\frac{\partial^2\psi}{\partial t^2}$$

establishes that the wave propagates at speed

$$\$v = c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}\$.$$

Worked Examples

Given the vector field \(vec(F) = (2xy + z^3)hati + (x^2 + 3y^2)hatj + (3xz^2)hatk\), determine whether \(vec(F)\) is conservative (irrotational) and find the divergence of \(vec(F)\) at the point \(P(1, -1, 2)\).

Step 1: Identify components of

$\vec{F}$:

$\$F_x = 2\$\$xy + z^3,$

\(

F_y = x^2 + 3

$y^2\), \($

F_z = 3

$xz^2\). Step 2 : Check if \(\vec{F}\) is irrotational by computing \($

nabla x vec(F)\):

$$\n\$abla \times \vec{F} = \hat{i}\$\left(\$\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\$\right) - \hat{j}\left(\$\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\$\right) + \hat{k}\left(\$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\$\right)$$

- \((partial Fz)/(partial y) = 0\), \((partial Fy)/(partial z) = 0 implies (0 - 0)hati = 0hati\)
- \((partial Fz)/(partial x) = 3z^2\), \((partial Fx)/(partial z) = 3z^2 implies -(3z^2 - 3z^2)hatj = 0hatj\)
- \((partial Fy)/(partial x) = 2x\), \((partial Fx)/(partial y) = 2x implies (2x - 2x)hatk = 0hatk\)

Since

$$\$\nabla \times \vec{F} = \vec{0}\$,$$

the vector field is conservative (irrotational). Step 3: Calculate the divergence

$$\$\nabla \cdot \vec{F}\$:$$

$$\n\$abla \cdot \vec{F} = \frac{\partial}{\partial x}\$(2\$xy + z^3) + \frac{\partial}{\partial y}\$(\$x^2 + 3\$\$y^2) + \frac{\partial}{\partial z}\$(3\$xz^2) = 2\$\$y + 6\$\$y + 6\$\$x$$

Step 4: Evaluate

$$\$\nabla \cdot \vec{F}\$$$

at point

$$P(1, -1, 2) :$$

$$\n\$abla \cdot \vec{F}\$$$

Final Answer: The vector field

$$\vec{F}$$

is conservative (irrotational) because

$$\$\nabla \times \vec{F} = \vec{0}\$.$$

The divergence at

$$(1, -1, 2)$$

is

$$4.$$

A parallel-plate capacitor with circular plates of radius \(R = 0.1 m\) is separated by air. The electric field between the plates changes at a uniform rate of \((dE)/(dt) = 1.0 x 10^(12)V/(mcdots)\). Calculate the displacement current density \(Jd\) and the total displacement current \(Id\) passing between the plates. (Take \(varepsilon_0 = 8.854 x 10^(-12) F/m\)).

Step 1: Identify given quantities:

- Plate radius \(R = 0.1 m\)
- Permittivity of air \(~ varepsilon_0 = 8.854 x 10^(-12) F/m\)
- Rate of change of electric field \((dE)/(dt) = 1.0 x 10^(12)V/(mcdots)\)

Step 2: Calculate displacement current density

J_d:

$$J_{\$}d = \varepsilon_0 \$ \frac{dE}{dt}$$

$$J_d = (\$8.854 \times 10\$^{-12}\ F/m) \times (\$1.0 \times 10\$^{12}\$\ \mathrm{V}/(\mathrm{m}\cdot\mathrm{s})\$) = 8.854\ A/m^2$$

Step 3: Calculate the area

A

of the circular plates:

$$A = \$\pi R^2 = \pi\$(0.1\ \mathrm{m})^{\$}2 = 0.01\$\pi\$\ \mathrm{m}^2 \approx 0.031416\$\ \mathrm{m}^2$$

Step 4: Compute the total displacement current

I_d:

$$I_{\$}d = J_d \times A = 8.854\$\$\ \mathrm{A}/\mathrm{m}^2 \times 0.01\$\pi\$\ \mathrm{m}^2 \approx 0.278\$\ \mathrm{A}$$

Final Answer: The displacement current density is

$8.854\ A/m^2$

and the total displacement current is

$0.278\ \mathrm{A}$

(or

$278\ \mathrm{mA}$

).

Applications

- Capacitive Circuits: Explains how alternating current (AC) effectively crosses the insulating dielectric gap of a capacitor via displacement current.
- Antenna Design and Wireless Transmission: Time-varying currents give rise to coupled, radiating electric and magnetic fields that propagate as radio waves.
- Electromagnetic Shielding (EMC): Understanding field divergence and curl helps electrical engineers design conductive enclosures that prevent high-frequency interference.
- Transmission Lines and Waveguides: Maxwell's equations define the boundary conditions that guide TEM, TE, and TM modes along co-axial cables and hollow metal guides.

Exam Questions with Answers

- State Gauss's Divergence Theorem. Use it to transform the differential form of Gauss's Law into its integral form.

Gauss's Divergence Theorem states that the volume integral of the divergence of a vector field

$\vec{D}$

over a closed volume

V

is equal to the total outward flux of

$\vec{D}$

through the enclosing surface

S:

$$\iiint_V (\$\nabla \cdot \vec{D}\$)\, \$dV = \oint_S \$\$\vec{D} \cdot d\$\vec{S}$$

From the differential form of Gauss's Law,

$$\$\nabla \cdot \vec{D} = \rho_v.$$

Integrating both sides over volume

V:

$$\iiint_V (\$\nabla \cdot \vec{D}\$)\, \$dV = \iiint_V \$\rho_v \, dV$$

By applying the divergence theorem on the left and recognizing that

on the right, we obtain the integral form:

- Why did Ampÿýre's circuital law require modification for time-varying fields? Show how Maxwell resolved this inconsistency.

Ampÿýre's original law is

$$\$\nabla \times \vec{H} = \vec{J}\$.$$

Taking the divergence gives

$$\nabla \cdot (\$\nabla \times \vec{H}\$) = \$\nabla \cdot \vec{J}\$.$$

The

$$left - hand$$

side is identically zero, implying

$$\$\nabla \cdot \vec{J} = 0\$.$$

However, conservation of electric charge requires the continuity equation

$$\$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}\$.$$

This created a contradiction for

$$non - steady$$

states where charge accumulates over time (

$$\$\frac{\partial \rho_v}{\partial t} \neq 0\$$$

). Maxwell resolved this by using Gauss's law

$$\$\rho_v = \nabla \cdot \vec{D}\$$$

inside the continuity equation, rewriting it as

He added

$$\frac{\partial \vec{D}}{\partial t}$$

(the displacement current density) to Ampÿýre's law:

$$\$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}\$.$$

Common Mistakes

- Treating the Del operator (?) as a simple vector instead of a differential operator: ? cannot be rearranged or commuted freely past variables.
- Confusing displacement current with conduction current: Conduction current involves physical transport of charge carriers (electrons or ions), whereas displacement current is the time rate of change of an electric displacement field without real charge transport.
- Forgetting the negative sign in Faraday's Law: \(nabla x vec(E) = -fracpartial vec(B)partial t\); the minus sign represents Lenz's Law (conservation of energy).
- Applying Gauss's Law to open surfaces: Gauss's divergence theorem and Gauss's law strictly require closed surfaces that completely enclose a 3D volume.

Practice Questions with Solutions

- Evaluate the divergence and curl of the magnetic vector field \(vec(B) = (2x)hati - (2y)hatj + 0hatk\). State whether this represents a physically possible static magnetic field.

Hint: Check whether

$$\$\nabla \cdot \vec{B} = 0\$$$

(Gauss's law for magnetism). Solution: Step 1: Calculate divergence

$$\$\nabla \cdot \vec{B} = \frac{\partial(2x)}{\partial x} + \frac{\partial(-2y)}{\partial y} + \frac{\partial(0)}{\partial z} = 2 - 2 + 0 = 0\$.$$

Step 2: Calculate curl

$$\$\nabla \times \vec{B} = \hat{i}\$(\$0 - 0\$) - \hat{j}(\$0 - 0\$) + \hat{k}\left(\$\frac{\partial(-2y)}{\partial x} - \frac{\partial(2x)}{\partial y}\$\right) = \vec{0}.$$

Step 3: Conclusion: Because

$$\$\nabla \cdot \vec{B} = 0\$,$$

it satisfies Maxwell's second equation and is physically realizable as a static magnetic field in a

$$current - free$$

region.

Quick Revision Summary

1. The del operator \(nabla\) generates the gradient (scalar to vector), divergence (vector to scalar), and curl (vector to vector).
2. Integral theorems (Divergence and Stokes') bridge localized field differentials to macroscopic closed-surface and closed-loop measurements.
3. Maxwell's four equations describe how spatial derivatives of fields are tied to static sources (charges) and temporal variations of adjacent fields.
4. Displacement current density \(vec(J)d = fracpartial vec(D)partial t\) ensures charge conservation and makes wave propagation possible in a vacuum with speed \(c = frac1" (mu0 varepsilon_0)\).